Throughout my education and career, I've moved across different domains, but consistently toward data-focused work. During my bachelor's, I realised I was more interested in data and statistics than traditional software development, which led me to pursue data science and AI in my master's at Uppsala. My experiences have ranged from building a database for an underfunded school to analysing network data for my thesis. Across these, I've leaned toward R&D-oriented roles, as I'm more motivated by exploring new possibilities than maintaining existing systems. While consultancy seemed like a natural path, the constant focus on delivering acceptable solutions within tight constraints limited the depth of work, which didn't suit how I prefer to approach problems.

Growing up in a hospital compound where my father worked gave me early exposure to healthcare systems. I volunteered there and saw firsthand how difficult it can be to standardise data and terminology, even within a single department. During my master's, I built on this interest through courses in healthcare innovation, covering the legal and business aspects of healthtech in Sweden, as well as a project with Akademiska Sjukhuset and a startup focused on interoperability challenges, EHR systems, and health data standards. These experiences have shaped my understanding of the Swedish healthcare landscape.

While exploring applied AI in healthcare, I was introduced to omics research and its role in cancer. Around the same time, a friend undergoing personalised cancer treatment in Sweden exposed me to the advances in this space. Proteomics stands out because it closely reflects what is happening in tumours while still requiring substantial computational analysis. There's a whole world of information and patterns to uncover, and it's why this PhD stood out to me. The opportunity to analyse omics data and reconstruct signalling networks that could inform treatment decisions. Before my grandmother passed away from breast cancer complications, we chose a palliative care approach, and she remained very active until her final days. Working towards better information can support better decisions for patients and families.

I don't have any formal education with omics data, but I just started a small pilot project with the Heme/Onc department at King Faisal Hospital in Saudi Arabia. The goal is to build an AI layer to assist in interpreting capillary electrophoresis outputs, which are currently reviewed manually by experts. I assume the PhDs' data will come from mass spectrometry, but I'm learning about reading graphs and spikes, and how AI researchers train for that. The project focuses on building models to classify abnormal patterns, localise peaks, and generate structured draft reports, to reduce turnaround time while maintaining clinical reliability.

On the purely technical side, I'm very comfortable with Python and its various ML libraries. I actually just put a finetuned LLM annotator to run for my thesis before I began writing this letter, with my one gap being that I still haven't had a chance to work on a RAG project.

The way this project combines computational methods with real biological questions is exactly what I am looking for. With the support of the multi-disciplinary team, I'm excited to work with everyone and am confident the project will lead to a positive impact.part of it.